

EP3120 - Statistical Mechanics

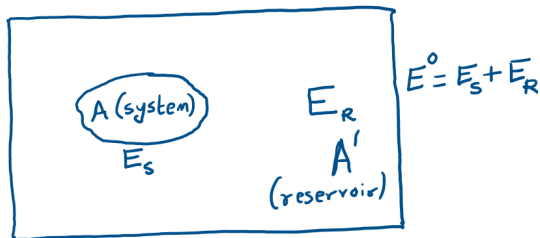
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Canonical Ensembles

Systems in thermal bath

- Consider a small system A in contact with a heat reservoir A' , with the system much smaller than the reservoir
- The reservoir is at a fixed temperature that is the same as temperature of the system
- There are many examples of this, like a bottle of beer in a swimming pool or a ladle of molten steel in a furnace
- Often times the system could also be microscopic, for ex. it could also be a bunch of molecules in a sea of molecules (in a gas), or a group of atoms in a lattice (in a solid)
- Let the system be in a microstate s with energy E_s
- Note: s is a label for microstate, not energy. There could be different microstates s with the same energy
- Let E_R be the energy of the reservoir

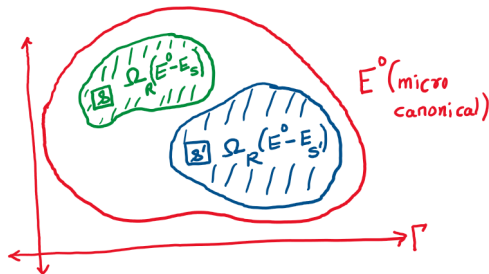
Canonical ensemble



- The total energy of the system+reservoir is constrained narrowly within $E^0, E^0 + \Delta$
- This means that the reservoir energy also lies very close to the value $E_R = E^0 - E_s$
- The microstates of this entire system+reservoir are part of a microcanonical ensemble
- What is the probability of finding the system in a particular state s within this microcanonical ensemble?

Probability of a microstate

- When the system is in a particular state s , the number of microstates of the system+reservoir is equal to the number of microstates of the reservoir with energy $E^0 - E_s$



- This means that the probability of finding the system in microstate s is proportional to $\Omega_R(E^0 - E_s)$

$$P_s = C \Omega_R(E^0 - E_s) \quad (1)$$

- The proportionality constant C can be found by $\sum_s P_s = 1$, where the sum is over all possible microstates (not just energies)

Probability distribution

- From Taylor series we have

$$\Omega_R(E^0 - E_s) = \Omega_R(E^0) - E_s \left. \frac{\partial \Omega_R}{\partial E} \right|_{E^0} + \frac{E_s^2}{2} \left. \frac{\partial^2 \Omega_R}{\partial E^2} \right|_{E^0} + \dots \quad (2)$$

- This can be simplified by making use of the thermodynamic relation

$$\frac{\partial S}{\partial E} = \frac{1}{T} \implies \frac{1}{\Omega} \frac{\partial \Omega}{\partial E} = \frac{1}{k_B T} \equiv \beta \quad (3)$$

- Similarly

$$\frac{\partial^2 \Omega}{\partial E^2} = \frac{\partial(\beta \Omega)}{\partial E} \approx \beta \frac{\partial \Omega}{\partial E} = \beta^2 \Omega \quad (4)$$

- Here we have used the fact the Ω is a much rapidly increasing function of E while β changes much slowly with E
- We have also used the fact that the temperature of the system and reservoir are the same T

Probability distribution

- Putting together we get

$$\Omega_R(E^0 - E_s) \approx \Omega_R(E^0) \left[1 - \beta E_s + \frac{(\beta E_s)^2}{2} + \dots \right] \quad (5)$$

$$= \Omega_R(E^0) \exp[-\beta E_s] \quad (6)$$

- The $\Omega_R(E^0)$ is a dimensionless constant and so we have

$$P_s = C e^{-\beta E_s} \quad (7)$$

- We get the canonical ensemble distribution function

$$P_s = \frac{e^{-\beta E_s}}{Q} \quad (8)$$

- Where Q is the canonical partition function

$$Q(N, V, T) = \sum_s e^{-\beta E_s} \quad (9)$$

- Since we are summing up over the energies, it is no longer a variable, instead we now have the temperature

Average energy and free energy

- The thermodynamic variable of internal energy, U , is the average energy of the system in this canonical ensemble

$$U \equiv \langle E \rangle = \sum_s P_s E_s = \frac{\sum_s E_s e^{-\beta E_s}}{\sum_s e^{-\beta E_s}} \quad (10)$$

$$= -\frac{\partial}{\partial \beta} \{ \ln [\sum_s e^{-\beta E_s}] \} = -\frac{\partial \ln Q}{\partial \beta} \quad (11)$$

- We can obtain the internal energy if we know the partition function
- The Helmholtz free energy ($A = U - TS$) obeys (for constant N)

$$dA = dU - TdS - SdT = -PdV - SdT \quad (12)$$

$$\Rightarrow S = -\left. \frac{\partial A}{\partial T} \right|_V \quad (13)$$

- Now let's obtain the relation between A and Q

Entropy

- Since $U = A + TS$, we obtain

$$U = A - T \left. \frac{\partial A}{\partial T} \right|_V = -T^2 \frac{\partial}{\partial T} \left(\frac{A}{T} \right)_V = \left[\frac{\partial(A/T)}{\partial(1/T)} \right]_V \quad (14)$$

$$= \frac{1}{k_B} \left[\frac{\partial(A/T)}{\partial \beta} \right]_V \quad (15)$$

- Comparing this with the expression obtained earlier,
 $U = -\partial(\ln Q)/\partial\beta$, we get the expression

$$\frac{A}{T} = -k_B \ln Q \implies A(N, V, T) = -k_B T \ln Q \quad (16)$$

- Now consider the quantity

$$\langle \ln(P_s) \rangle = \sum_s P_s \ln(P_s) \quad (17)$$

Entropy

- Substitute the relation $P_s = e^{-\beta E_s}/Q$

$$\langle \ln(P_s) \rangle = \sum_s \frac{e^{-\beta E_s}}{Q} \ln \left(\frac{e^{-\beta E_s}}{Q} \right) = \sum_s \frac{e^{-\beta E_s}}{Q} [-\beta E_s - \ln Q] \quad (18)$$

$$= -\beta \sum_s P_s E_s - \ln Q \quad (19)$$

$$= -\beta U + \beta A = \beta(A - U) = -\beta TS \quad (20)$$

- This gives a new definition of entropy

$$S = -k_B \langle \ln P_s \rangle = -k_B \sum_s P_s \ln P_s \quad (21)$$

- The probability distribution of the different micro-states solely determines the entropy of the system
- This motivates the information theory entropy where the entropy of information coming in some discrete variable X_i with probability p_i is

$$S_{Shannon} = - \sum_i p_i \ln p_i \quad (22)$$

Entropy

- H.W.: Prove that the entropy is maximized in a micro-canonical ensemble and it reduces to the Boltzmann formula
- If we are considering a system with energy levels E_i (here i represents the energy level, not the microstate)
- Let there be g_i number of microstates with energy E_i , i.e. a degeneracy of g_i
- Then the partition function can be written as

$$Q = \sum_i g_i e^{-\beta E_i} \quad (23)$$

- Here the summation is over energy levels, not microstates
- If the energy is a continuous variable then

$$Q = \int dE g(E) e^{-\beta E} \quad (24)$$

- Where $g(E)$ is the density of states and $g(E)dE$ is the number of states between energies E and $E + dE$

Partition function of classical ideal gas

- The energy of a particular microstate (with N particles) is

$$E_s = \sum_{i=1}^N \frac{p_i^2}{2m} \quad (25)$$

- The different states s correspond to different combinations of p_i
- Now the energy is not constrained to within E and $E + \Delta$, and so we can allow all the microstates in Γ -space
- As before we will coarse grain the Γ -space so that one microstate corresponds to h^{3N} volume
- The partition function can then be converted from a discrete sum to an integral

$$Q(N, V, T) = \sum_s e^{-\beta E_s} = \int \frac{d^N \mathbf{r} d^N \mathbf{p}}{h^{3N}} \exp\left\{-\beta \sum_{i=1}^N \frac{p_i^2}{2m}\right\} \quad (26)$$

- Exercise: Calculate the partition function

Partition function of classical ideal gas

- We should also add the Gibbs correction and divide this by $N!$

$$Q = \frac{V^N}{N! h^{3N}} \int d\mathbf{p}_1 d\mathbf{p}_2 \dots d\mathbf{p}_N \prod_{i=1}^N \exp\left[-\beta \frac{p_i^2}{2m}\right] \quad (27)$$

$$= \frac{V^N}{N! h^{3N}} \left[\int d\mathbf{p} \exp\left(-\frac{\beta p^2}{2m}\right) \right]^N \quad (28)$$

$$= \frac{V^N}{N! h^{3N}} \left[\int_0^\infty dp 4\pi p^2 \exp\left(-\frac{\beta p^2}{2m}\right) \right]^N \quad (29)$$

$$= \frac{V^N}{N! h^{3N}} \left[2m\pi k_B T \right]^{3N/2} \quad (30)$$

- This can also be expressed as, where Q_1 is the single particle partition function,

$$Q = \frac{1}{N!} Q_1^N; \quad Q_1 = \left[\frac{1}{h^3} \int d\mathbf{r} d\mathbf{p} \exp\left(-\frac{\beta p^2}{2m}\right) \right]^N \quad (31)$$

Free energy and entropy

- When the particles are not interacting, then the partition function of the whole system can be deduced from the partition function of a single particle
- The free energy becomes

$$A(N, V, T) = -k_B T \ln Q \quad (32)$$

$$= Nk_B T \left[\ln \left\{ \frac{N}{V} \left(\frac{h^2}{2\pi m k_B T} \right)^{3/2} \right\} - 1 \right] \quad (33)$$

- Here the Stirling approximation is used
- Once again, we can derive the entire thermodynamics from this expression. For ex.

$$P = - \left. \frac{\partial A}{\partial V} \right|_{N, T} = \frac{Nk_B T}{V} \quad (34)$$

Entropy

- We can now obtain the entropy as follows

$$S = -\frac{\partial A}{\partial T} \Big|_{N,V} \quad (35)$$

$$= Nk_B \left\{ \ln \left[\frac{V}{N} \left(\frac{2\pi mk_B T}{h^2} \right)^{3/2} \right] + \frac{5}{2} \right\} \quad (36)$$

- Earlier we had derived the Sackur-Tetrode equation for entropy of an ideal gas using the micro-canonical ensemble

$$S = Nk_B \ln \left[\frac{V}{N} \left(\frac{4\pi m E}{h^2 3N} \right)^{3/2} \right] + \frac{5}{2} Nk_B \quad (37)$$

- If we substitute the energy $E = (3/2)Nk_B T$ in the Sackur-Tetrode expression, we recover exactly the same entropy as derived from the canonical ensemble approach
- However, the advantage of canonical ensemble approach is that its calculation is easier

Connection between microcanonical and canonical

- Why do we get the same results from a canonical approach?
- Lets calculate the energy fluctuations in an ideal gas in a canonical ensemble

$$\Delta E = E - \langle E \rangle \quad (38)$$

- The rms fluctuation level is

$$\Delta E_{rms} = \sqrt{\langle (\Delta E)^2 \rangle} \quad (39)$$

$$= \sqrt{\langle E^2 - 2E\langle E \rangle + \langle E \rangle^2 \rangle} \quad (40)$$

$$= \sqrt{\langle E^2 \rangle - \langle E \rangle^2} \quad (41)$$

- We already know that $\langle E \rangle = U = (3/2)Nk_B T$. We calculate

$$\langle E^2 \rangle = \sum_s P_s E_s^2 \quad (42)$$

Energy fluctuations

- This becomes

$$\langle E^2 \rangle = \sum_s E_s^2 \frac{e^{-\beta E_s}}{Q} = \frac{1}{Q} \frac{\partial^2 Q}{\partial \beta^2} \quad (43)$$

- Making use of the expression of Q already derived, we obtain

$$\langle E^2 \rangle = \left(\frac{3N}{2} \right) \left(\frac{3N}{2} + 1 \right) (k_B T)^2 = \left(\frac{3}{2} N k_B T \right)^2 + \frac{3N}{2} (k_B T)^2 \quad (44)$$

$$\Rightarrow \Delta E_{rms} = \sqrt{\frac{3N}{2}} (k_B T) \Rightarrow \frac{\Delta E_{rms}}{\langle E \rangle} \sim \frac{1}{\sqrt{N}} \quad (45)$$

- We see that the fluctuation levels in a canonical ensemble are of the same order as the error Δ we had taken in a microcanonical ensemble
- This means that the vast majority of microstates in the canonical ensemble are similar to the states in the microcanonical ensemble

Equipartition theorem

- Consider a general system described by some generalized co-ordinates $q_i, p_i, i = 1, \dots, N$
- The Hamiltonian (energy) of this system is given by

$$H = \sum_{i=1}^N A_i p_i^2 + \sum_{i=1}^N B_i q_i^2 \quad (46)$$

where A_i and B_i are some constant co-efficients

- An example would be collection of classical simple harmonic oscillators in which case $A_i = (1/2m)$ and $B_i = (k/2)$ with spring constant k
- The equations of motion are

$$\dot{p}_i = -\frac{\partial H}{\partial q_i} \quad (47)$$

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \quad (48)$$

Equipartition theorem

- Consider a canonical ensemble of this system, so the probability of a particular state is

$$P_s \propto e^{-\beta H_s} \quad (49)$$

- Lets calculate the following

$$\left\langle q_i \frac{\partial H}{\partial q_j} \right\rangle = \frac{\int d\omega q_i \frac{\partial H}{\partial q_j} e^{-\beta H}}{\int d\omega e^{-\beta H}} \quad (50)$$

where $d\omega = dp_1 dp_2 \dots dp_N dq_1 \dots dq_N$

- The integral over q_j in the numerator becomes

$$\int dq_j q_i \frac{\partial H}{\partial q_j} e^{-\beta H} = -\frac{1}{\beta} \int dq_j q_i \frac{\partial}{\partial q_j} e^{-\beta H} \quad (51)$$

Equipartition theorem

- Doing the integral by parts

$$\int dq_j q_i \frac{\partial H}{\partial q_j} e^{-\beta H} = -\frac{1}{\beta} \left[q_i e^{-\beta H} \right]_{q_j^{\min}}^{q_j^{\max}} + \frac{1}{\beta} \int dq_j \delta_{ij} e^{-\beta H} \quad (52)$$

- The first term will be zero because $H \rightarrow \infty$ at extreme values of the co-ordinates
- Substituting back gives

$$\left\langle q_i \frac{\partial H}{\partial q_j} \right\rangle = \frac{\delta_{ij}}{\beta} \Rightarrow \langle B_i q_i^2 \rangle = \frac{k_B T}{2} \quad (53)$$

- Similarly we will get

$$\langle A_i p_i^2 \rangle = \frac{k_B T}{2} \quad (54)$$

- This is the equipartition theorem which states that for a harmonic Hamiltonian each degree of freedom contains on an average $(1/2)k_B T$ of energy

Simple Harmonic Oscillators

- Consider a system of N classical simple harmonic oscillators
- They are 1-D with energy given by

$$E_i = \frac{1}{2}m\omega^2 q_i^2 + \frac{1}{2m}p_i^2; \quad i = 1, 2, 3, \dots, N \quad (55)$$

- The one-particle partition function is

$$Q_1 = \frac{1}{h} \int_{-\infty}^{\infty} dq \int_{-\infty}^{\infty} dp e^{-\beta E} \quad (56)$$

$$= \frac{1}{h} \left[\int_{-\infty}^{\infty} dq e^{-\beta m\omega^2 q^2/2} \right] \left[\int_{-\infty}^{\infty} dp e^{-\beta p^2/2m} \right] \quad (57)$$

- Use $\int_{-\infty}^{\infty} dx e^{-\alpha x^2} = \sqrt{\pi/\alpha}$

$$Q_1 = \frac{1}{h} \left[\sqrt{\frac{2\pi}{\beta m\omega^2}} \right] \left[\sqrt{\frac{2\pi m}{\beta}} \right] = \frac{2\pi}{h\beta\omega} = (\beta\hbar\omega)^{-1} \quad (58)$$

Classical SHM partition function

- Then the N -particle partition function becomes

$$Q = Q_1^N = (\beta \hbar \omega)^{-N} \quad (59)$$

- Here we are not taking the Gibbs correction of dividing by $N!$
- This is because in realistic problems, these oscillators do not represent particles, but energy levels of particles
- Ex. energy levels of light (photons) or of lattice vibrations (phonons)
- These energy levels are distinguishable, and so Gibbs correction is not added here
- The free energy becomes

$$A(N, T) = -k_B T \ln Q \quad (60)$$

$$= Nk_B T \ln\left(\frac{\hbar \omega}{k_B T}\right) \quad (61)$$

Thermodynamic relations

- From Maxwell relations we have

$$P = -\frac{\partial A}{\partial V} = 0 \quad (62)$$

$$S = -\frac{\partial A}{\partial T} = Nk_B \left[1 + \ln \left(\frac{k_B T}{\hbar \omega} \right) \right] \quad (63)$$

$$U = -\frac{1}{Q} \frac{\partial Q}{\partial \beta} = Nk_B T \quad (64)$$

$$H = U \quad (65)$$

$$C_P = C_V = \frac{\partial U}{\partial T} = Nk_B \quad (66)$$

- The internal energy obeys the equipartition theorem

Quantum SHM

- Consider the SHM problem in the case where the oscillator obeys the laws of quantum mechanics
- From QM it is found that the energy of this oscillator is given by the form

$$E = \left(n + \frac{1}{2}\right) \hbar\omega; \quad n = 0, 1, 2, \dots \quad (67)$$

- This shows that the energy of quantum SHM is quantized in steps of $\hbar\omega$
- The lowest energy state possible is $\hbar\omega/2$ - ground state energy
- Consider a microcanonical ensemble of N quantum SHMs
- Since energy is quantized, we can consider a microcanonical ensemble with $\Delta < \hbar\omega$
- This implies that the energy of each microstate in the ensemble is exactly equal to some value E

Microcanonical Quantum SHM

- The total energy has to be half integer multiple of $\hbar\omega$
- Let's redefine the energy of each oscillator

$$E_i = \left(k_i - \frac{1}{2}\right)\hbar\omega; \quad k_i = 1, 2, 3, \dots \quad (68)$$

- The total energy of N oscillators becomes

$$E = \sum_{i=1}^N \left(k_i - \frac{1}{2}\right)\hbar\omega \equiv \left(K - \frac{N}{2}\right)\hbar\omega \quad (69)$$

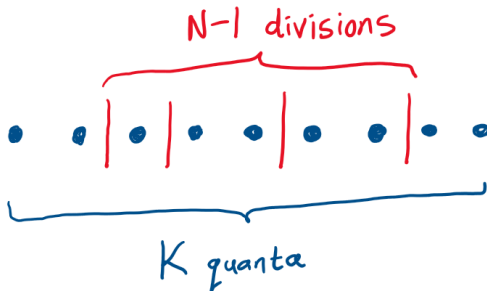
- Here

$$K = \sum_{i=1}^N k_i = \frac{E}{\hbar\omega} + \frac{N}{2} \quad (70)$$

$$K \geq N \quad (71)$$

Number of microstates

- The energy is fixed and so the K is also fixed
- In order to count the number of microstates in this ensemble, we need to enumerate the number of ways these K quanta can be distributed among N oscillators, such that each oscillator receives at least one quantum
- Imagine these quanta arranged in a line
- Each microstate corresponds to a particular way of dividing them up into N partitions - in other words putting $N - 1$ dividing lines between them



Number of microstates

- The quanta are indistinguishable, however the partitions (or the oscillators) are distinguishable
- The number of ways of placing $N - 1$ partitions in $K - 1$ spaces is

$$\Omega = {}^{K-1}C_{N-1} = \frac{(K-1)!}{(N-1)!(K-N)!} \quad (72)$$

- The entropy then becomes

$$S = k_B \ln \Omega \quad (73)$$

$$= k_B [\ln(K-1)! - \ln(N-1)! - \ln(K-N)!] \quad (74)$$

$$= k_B [(K-1) \ln(K-1) - (N-1) \ln(N-1) - (K-N) \ln(K-N)] \quad (75)$$

- In the classical SHM case, the energy per oscillator would be much bigger than a single quanta

$$K \gg N \gg 1; \quad K = \frac{E}{\hbar\omega} + \frac{N}{2} \approx \frac{E}{\hbar\omega} \quad (76)$$

Entropy of microcanonical SHM

- Then the entropy reduces to

$$S \approx k_B \ln \left[K \ln(K) - N \ln N - (K - N) \ln(K - N) \right] \quad (77)$$

- Using Taylor series expansion for the last term

$$\ln(K - N) = \ln(K) + \ln \left[1 - \frac{N}{K} \right] \approx \ln(K) - \frac{N}{K} \quad (78)$$

- Substituting in the entropy expression

$$S \approx k_B \left[-N \ln N + N \ln K + (K - N) \frac{N}{K} \right] \quad (79)$$

$$\approx Nk_B \ln \left[\frac{K}{N} \right] + Nk_B \quad (80)$$

$$= Nk_B \left[1 + \ln \left(\frac{E}{N\hbar\omega} \right) \right] \quad (81)$$

Temperature

- We can define the temperature of this microcanonical ensemble

$$\frac{1}{T} = \left. \frac{\partial S}{\partial E} \right|_N = \frac{Nk_B}{E} \quad (82)$$

- Substituting this back in the entropy expression we get

$$S = Nk_B \left[1 + \ln \left(\frac{k_B T}{\hbar \omega} \right) \right] \quad (83)$$

- This is the same expression as was obtained for the classical SHM in a canonical ensemble
- Here we have recovered it by taking the classical limit ($E/N \gg \hbar \omega$) of a microcanonical ensemble of quantum SHMs
- Exercise: Repeat the calculations for a canonical ensemble of quantum SHMs. Start by calculating the partition function

Paramagnetism

- Paramagnetic substances are weakly magnetized in an external magnetic field and attracted weakly to magnets, ex. nickel, aluminum, sodium
- Consider a system of N magnetic dipoles with dipole moments μ_i
- The magnitude of these dipole moments are the same for all the dipoles $\mu = |\mu_i|$
- Assume that these dipoles are static and non-interacting with each other
- In the presence of an external magnetic field $\mathbf{B}$ pointing in the z direction, these dipoles have a potential energy

$$E_i = -\mu_i \cdot \mathbf{B} = -\mu B \cos \theta_i \quad (84)$$

- Here θ_i is the angle the dipole makes with the magnetic field
- The total energy of the system becomes

$$E = \sum_{i=1}^N E_i = -\mu B \sum_i \cos \theta_i \quad (85)$$

Canonical ensemble

- Now consider a canonical ensemble of this system at some temperature T
- The different microstates in this ensemble correspond to different orientations of the dipoles with respect to the magnetic field
- The partition function would be

$$Q = Q_1^N = \left[\sum_{\text{orientations}} e^{-\beta E} \right]^N \quad (86)$$

- Get the single particle partition function

$$Q_1 = \sum_{\text{orientations}} e^{\beta \mu B \cos \theta} \quad (87)$$

Classical dipole

- If this is a classical dipole, then it can take any orientation with respect to the magnetic field
- Then we integrate over all possible orientations to obtain the single particle partition function

$$Q_1 = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta e^{\beta \mu B \cos \theta} \quad (88)$$

$$= 4\pi \frac{\sinh(\beta \mu B)}{\beta \mu B} \quad (89)$$

- The N particle partition function becomes

$$Q = Q_1^N = \left[\frac{4\pi \sinh(\beta \mu B)}{\beta \mu B} \right]^N \quad (90)$$

- The different lattice sites are distinguishable

Average magnetization

- The mean magnetic moment will have to be in the direction of the magnetic field (z-direction)
- The mean magnetic moment will be given by

$$\bar{M} = N\mu\langle\cos\theta\rangle \quad (91)$$

- The average of $\cos\theta$ can be obtained from the partition function

$$\langle\cos\theta\rangle = \frac{\int d\theta_1 \dots d\theta_N \sin\theta_1 \dots \sin\theta_N (\sum_i \cos\theta_i) e^{\beta\mu B \sum_j (\cos\theta_j)}}{N \int d\theta_1 \dots d\theta_N \sin\theta_1 \dots \sin\theta_N e^{\beta\mu B \sum_j \cos\theta_j}} \quad (92)$$

$$= \frac{\int d\theta \sin\theta \cos\theta e^{\beta\mu B \cos\theta}}{\int d\theta \sin\theta e^{\beta\mu B \cos\theta}} \quad (93)$$

$$= \frac{1}{\beta\mu Q_1} \frac{\partial Q_1}{\partial B} \quad (94)$$

- The ensemble average reduces to a single particle average because the particles are not interacting

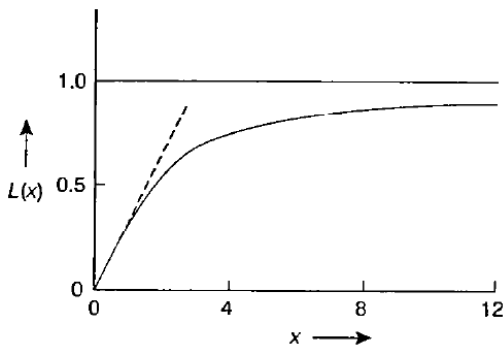
Average magnetization

- Substituting the expression for Q_1 gives

$$\langle \cos \theta \rangle = \coth(\beta\mu B) - \frac{1}{\beta\mu B} \quad (95)$$

- We defined the Langevin function

$$L(x) \equiv \coth(x) - \frac{1}{x} \quad (96)$$



Curie law of paramagnetism

- The mean magnetic moment then becomes

$$\bar{M} = N\mu L(\beta\mu B) \quad (97)$$

- In the limit of strong B , i.e., $\beta\mu B \gg 1$, $L(x) \Rightarrow 1$
- Then the mean magnetic moment becomes $N\mu$
- For high temperatures or weak magnetic fields $\beta\mu B \ll 1$
- In this limit the Langevin function becomes

$$L(x) \approx \frac{x}{3} - \frac{x^3}{45}; \quad |x| \ll 1 \quad (98)$$

- In this limit the magnetization obeys

$$\bar{M} \approx \frac{N\mu^2 B}{3k_B T} \quad (99)$$

Experimental verification of Curie law

- This can be expressed as Curie's law of paramagnetism

$$\bar{M} = \frac{NCB}{T} \quad (100)$$

- C is Curie's constant
- This is experimentally verified by measuring the magnetic susceptibility χ

$$\chi = \frac{\bar{M}}{NB} = \frac{C}{T} \quad (101)$$

- The Curie law has been verified in various substances by measuring their magnetization under weak applied field

